A position-weighted refinement for simple zeros of the Riemann zeta function

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Abstract

We prove the unconditional lower bound

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq \frac{655000H_{\mathrm{MT}} - 1305}{652504} = 0.6730732086087052768351\ldots,$$

where

$$H_{\mathrm{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}} = 0.6725007036794116\ldots.$$

The new ingredient is a position-weighted refinement of the seven-point Gram-stability argument appearing in recent reproducible work. Earlier certificates use a uniform pressure on the six consecutive gaps. We observe that the global double-counting argument depends only on the total pressure, whereas the local extremal problem depends on its distribution. We exploit this freedom using the certified vector

$$\frac{1}{10^7}(2714, 3733, 3553, 3553, 3733, 2714), \qquad \sum_{j=1}^{6} p_j = \frac{1}{500},$$

and an Arb/FLINT interval certificate proving the corresponding seven-point lower bound 39/10000. Combined with block length $m = 262$, this yields the constant above.

For logical independence we also reconstruct the analytic/Gabor input package from classical ingredients rather than invoking the recent Theorem D as a black box. The external analytic trust base consists of the Guinand–Weil explicit formula, Stirling's formula, the Prime Number Theorem, the Riemann–von Mangoldt formula and standard local zero counting, standard prime-power estimates, the weighted Montgomery–Vaughan Hilbert inequality, and Poisson summation. The finite certificate is verified by directed interval arithmetic.

1 Introduction and statement of novelty

Let $N(T,2T)$ denote the number of nontrivial zeros of $\zeta(s)$ with ordinary ordinate in $(T,2T]$, counted with multiplicity, and let $N_0^s(T,2T)$ denote the number of simple zeros on the critical line in the same interval. The quantitative study of N_0^s/N goes back to Montgomery's pair-correlation method and the Montgomery–Taylor optimization. Recent work has made the relevant finite-compression and zero-side inertia mechanism unconditional and has produced reproducible improvements beyond the Montgomery–Taylor constant; see Claude [5] for the immediate comparison point and Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh [6, 7] and Goldston–Suriajaya [8, 9] for surrounding pair-correlation work. For the optimized Montgomery–Taylor test family we write

$$H_{\mathrm{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}} = 0.6725007036794116457\ldots. \tag{1}$$

Two very recent public computational manuscripts are especially close to the present argument. Jo's reproducible draft [13] obtains

$$0.6730085279277797613\ldots$$

using a stability-enhanced rank–trace inequality, a seven-point local functional, and a uniform pressure on its six gaps. A subsequent repository maintained by Rademacher [14], explicitly based on Jo's public artifact, strengthens the uniform seven-point certificate and obtains

$$0.6730213619501665335\ldots.$$

Jo's repository identifies its research draft as generated by GPT-5.6 Sol. Rademacher's repository records Jo's artifact as its upstream provenance and presents the subsequent refinement as AI-generated. We cite the human-maintained artifacts and treat model use as provenance rather than mathematical authorship.

The contribution of the present paper is narrower and explicit. We do not claim the stability-enhanced rank–trace inequality, the seven-point architecture, or the uniform-pressure local-to-global argument as new. Instead we optimize a degree of freedom left fixed in those works. If $g_1, \ldots, g_6$ are six consecutive normalized gaps, the global double-counting cost depends only on

$$\beta := \sum_{j=1}^{6} p_j,$$

whereas the local extremal problem depends on the individual p_j. Consequently the uniform choice can be replaced by a nonuniform position-dependent pressure without increasing the global cost. The certified choice used here is

$$p = \frac{1}{10^7}(2714, 3733, 3553, 3553, 3733, 2714), \qquad \sum_{j=1}^{6} p_j = \frac{1}{500}. \tag{2}$$

For this vector we certify

$$\mathcal{F}_p(g_1, \ldots, g_6) \geq \frac{39}{10000} \qquad (g_j \geq 0). \tag{3}$$

After global redistribution and shifted-block averaging, the block length $m = 262$ yields the theorem below.

Theorem 1.1 (Main theorem)**.** *With*

$$H_{\mathrm{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}},$$

one has unconditionally

$$\liminf_{T\to\infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{655000 H_{\mathrm{MT}} - 1305}{652504} = 0.6730732086087052768351\ldots.$$

A second contribution is expository and logical rather than a separate priority claim: we reconstruct the analytic/Gabor package needed by the proof from classical theorems and explicitly identify the common finite compression on its zero and prime sides. This makes the proof independent of using the recent Theorem D as an imported black box.

Priority convention. The priority statements above concern the public artifacts inspected while preparing this version. Jo's repository is the public upstream artifact for the stability/seven-point draft used here; Rademacher's repository records that provenance and strengthens its finite certificate. The present novelty claim is restricted to the nonuniform position-weighted refinement, its certified pressure vector, and the resulting constant.

2 The common finite compression and its two representations

Put
$$\ell = \log \frac{T}{2\pi}, \quad L = \ell, \quad h = \frac{2\pi}{L}, \quad d = \left\lfloor \frac{T}{h} \right\rfloor, \quad \tau_j = T + jh \quad (0 \le j < d).$$
Let $\phi = \phi_L \in C_c^2(\mathbb{R})$ be the real, even Montgomery–Taylor window supported on $[-L/2, L/2]$, and define
$$\widehat{\phi}(z) := \int_{\mathbb{R}} \phi(u)e^{-izu}\, du, \qquad \mathcal{F}_+ f(z) := \int_{\mathbb{R}} f(u)e^{izu}\, du.$$
Set
$$f_j(u) := \phi(u)e^{-i\tau_j u}. \tag{4}$$
Since ϕ is even, $\widehat{\phi}$ is even entire, hence
$$F_j(z) := \mathcal{F}_+ f_j(z) = \widehat{\phi}(\tau_j - z) = \widehat{\phi}(z - \tau_j). \tag{5}$$

For a zero $\rho = \beta + i\gamma$ put
$$z_\rho := -i\left(\rho - \frac{1}{2}\right) = \gamma - i\left(\beta - \frac{1}{2}\right). \tag{6}$$
For the horizontal partner $\rho^\sharp := 1 - \overline{\rho}$,
$$z_{\rho^\sharp} = \overline{z_\rho}. \tag{7}$$

Let W denote the Hermitian Weil form in the normalization of Appendix C. Define the single common finite compression
$$G_{jk} := W(f_j, f_k), \qquad 0 \le j, k < d. \tag{8}$$
For $c = (c_0, \ldots, c_{d-1})^t$, put
$$f_c := \sum_{j=0}^{d-1} \overline{c_j}\, f_j, \qquad \Lambda_\rho(c) := \mathcal{F}_+ f_c(z_\rho) = \sum_{j=0}^{d-1} \overline{c_j}\, F_j(z_\rho).$$
With this convention, which is chosen only to match the standard matrix quadratic form, one has $c^*Gc = W(f_c, f_c)$ exactly.

Proposition 2.1 (Two representations of the common compression)**.** *Let $X = e^L = T/(2\pi)$ and let $\nu_X = \mu + P_X + \Pi_X$ be the density derived from the classical Weil explicit formula in Proposition C.1. Then:*

(a) Spectral representation.
$$G_{jk} = \int_{\mathbb{R}} \widehat{\phi}(\tau - \tau_j)\overline{\widehat{\phi}(\tau - \tau_k)}\nu_X(\tau)\, d\tau. \tag{9}$$

(b) Zero-side representation. *For every $c \in \mathbb{C}^d$,*

$$c^*Gc = \sum_{\rho} m_\rho \, \Lambda_\rho(c) \, \overline{\Lambda_{\rho^\sharp}(c)}. \tag{10}$$

Grouping under $\rho \leftrightarrow \rho^\sharp$ gives

$$\begin{aligned} c^*Gc = &\sum_{\rho=\rho^\sharp} m_\rho |\Lambda_\rho(c)|^2 \\ &+ \sum_{\{\rho,\rho^\sharp\},\, \rho\neq\rho^\sharp} 2m_\rho \Re\Big(\Lambda_\rho(c)\overline{\Lambda_{\rho^\sharp}(c)}\Big)\,. \end{aligned} \tag{11}$$

(c) Interior/exterior decomposition. *Let $I' = (T - T^{1/2}, 2T + T^{1/2}]$. Define A by restricting the zero-side sum to zeros with ordinary ordinates in I', and E by the complementary sum. Since $\rho \mapsto \rho^\sharp$ preserves ordinary ordinate,*

$$G = A + E \tag{12}$$

entrywise.

(d) Normalized evaluation vectors. *Put*

$$a = \frac{1}{L}\int_{\mathbb{R}} \phi(u)^2 \, du, \qquad \widehat{A} = \frac{A}{aL^2}.$$

For a simple critical-line zero ρ in I', define

$$v_\rho := \frac{1}{\sqrt{a}\,L}\big(F_0(\gamma), \ldots, F_{d-1}(\gamma)\big)^t.$$

Then $\Lambda_\rho(c)/(\sqrt{a}\,L) = c^*v_\rho$, and

$$\|v_\rho\|^2 = \frac{1}{aL^2}\sum_{j=0}^{d-1} |\widehat{\phi}(\gamma - \tau_j)|^2 \leq 1. \tag{13}$$

Thus

$$P_1 := \sum_{\rho\in\mathcal{S}_1} v_\rho v_\rho^*$$

is exactly the normalized simple-critical-line part of $\widehat{A}$, and $Q' := \widehat{A} - P_1$ is the pull-back of the remaining zero-side blocks.

Proof. Proposition C.1 and (5) give (9). The zero-side form of the same Weil identity is

$$W(f,g) = \sum_\rho m_\rho \, \mathcal{F}_+ f(z_\rho) \, \overline{\mathcal{F}_+ g(\overline{z_\rho})}.$$

By the coefficient convention above, $c^*Gc = W(f_c, f_c)$. Using (7) with $f = g = f_c$ therefore proves (10); grouping paired zeros proves (11). Splitting this same zero sum by whether the ordinary ordinate lies in I' proves (12).

For a critical-line zero, $z_\rho = \gamma \in \mathbb{R}$. Extending the finite sum in (13) to the full translated critical lattice and applying Lemma D.2 at equal arguments (translation of the lattice changes only nonzero dual phases, whose alias terms vanish by exact support) gives

$$\sum_{j\in\mathbb{Z}} |\widehat{\phi}(\gamma - \tau_j)|^2 = aL^2.$$

Hence the finite sum is at most aL^2. Finally, (11) identifies the simple part with $\sum v_\rho v_\rho^*$ and the remainder with the pull-back of the multiple and off-line blocks. □

Remark 2.2 (Dependency graph)**.** *Input I uses the zero-side representation above, Input II uses $G = A + E$, Input III uses the spectral representation, and Input IV supplies the critical-lattice identity used in* (13)*. Thus all four modules refer to one test family and one matrix.*

3 Analytic/Gabor package and the stability refinement

Put
$$\ell := \log \frac{T}{2\pi}, \qquad I = (T, 2T], \qquad D_0 = T^{1/2}, \qquad I' = (T - D_0, 2T + D_0],$$
and set $L = \ell$. Let ϕ be the real, even, compactly tapered Montgomery–Taylor window used below, and define
$$a := \frac{1}{L}\int_{\mathbb{R}} \phi(u)^2\,du, \qquad \widehat{G} := \frac{G}{aL^2}, \qquad \widehat{A} := \frac{A}{aL^2}.$$

The analytic work needed by the stability argument is summarized in the next proposition. Its four assertions are proved independently in Appendices A–D.

Proposition 3.1 (Analytic/Gabor package)**.** *Let the zeros in I' be partitioned into simple critical-line zeros $\mathcal{S}_1$, distinct multiple critical-line zeros $\mathcal{S}_2$, and unordered horizontal pairs*
$$\{\rho, \rho^\sharp\}, \qquad \rho^\sharp = 1 - \overline{\rho},$$
off the critical line. Write $s_1 = \#\mathcal{S}_1$, $s_2 = \#\mathcal{S}_2$ and $p = \#\mathcal{P}$. Then the following hold.

(I) Zero-side inertia.
$$N(I') \geq s_1 + 2s_2 + 2p. \tag{14}$$
For the normalized sampled vectors v_ρ associated with $\rho \in \mathcal{S}_1$, let V have these vectors as columns and put
$$P_1 = VV^*, \qquad M = V^*V, \qquad Q' = \widehat{A} - P_1.$$
Then
$$n_+(Q') \leq s_2 + p. \tag{15}$$
(II) Exterior-zero tail. *Writing $\widehat{A} = \widehat{G} - \widehat{E}$,*
$$4\,\mathrm{tr}\,\widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 \geq 4\,\mathrm{tr}\,\widehat{G} - \|\widehat{G}\|_{\mathrm{F}}^2 - o(N(T, 2T)). \tag{16}$$
In fact the proof gives $\|\widehat{E}\|_1 \ll T^{-1/2}$.

(III) Optimized first and second moments.
$$\mathrm{tr}\,\widehat{G} = N(T, 2T)(1 + o(1)), \qquad \|\widehat{G}\|_{\mathrm{F}}^2 = \left(\frac{1}{c_1^*} + o(1)\right) N(T, 2T), \tag{17}$$
where
$$\frac{1}{c_1^*} = \frac{1}{2} + \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}}, \qquad 2 - \frac{1}{c_1^*} = H_{\mathrm{MT}}. \tag{18}$$
(IV) Critical-lattice Poisson–Gabor limit. *At mesh $h = 2\pi/L$, Poisson summation gives the infinite-grid identity without nonzero aliases. The tapered Fourier decay controls finite-grid truncation, and the normalized overlap kernel converges to the compact Montgomery–Taylor kernel used in Section 4. The integrated second-moment end effect is*
$$O\Big(L\ell\log L\,(\ell^2 + X)\Big), \qquad X = e^L \asymp T,$$
which is $o(TL\ell^2)$ at $L = \ell$.

Proof. Assertion (I) is Theorem A.5; assertion (II) is Theorem B.6; assertion (III) is proved in Appendix C, culminating in (83); and assertion (IV) is Theorem D.1. These appendices use only the classical trust base listed in Section 9. □

Define the convex function

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \le t \le 2, \\ 2t-3, & t \ge 2, \end{cases} \qquad \mathcal{D}(M) := \operatorname{tr} \Psi(M) \tag{19}$$

for $M \succeq 0$.

Lemma 3.2 (Stability-enhanced rank–trace inequality)**.** *Let V be a $d \times r$ matrix whose columns have norm at most one. Put $P = VV^* \succeq 0$ and $M = V^*V$. If Q is Hermitian and $n_+(Q) \le b$, then*

$$\|P+Q\|_{\mathrm{F}}^2 \ge 4\operatorname{tr}(P+Q) - 3r - 4b + \mathcal{D}(M). \tag{20}$$

Proof. Write $Q = Q_+ - Q_-$ for its positive and negative parts. They are positive semidefinite, have orthogonal supports, and $\operatorname{rank} Q_+ \le b$. Since $\operatorname{tr}((P-Q_-)Q_+) = \operatorname{tr}(PQ_+) \ge 0$,

$$\|P+Q\|_{\mathrm{F}}^2 \ge \|P-Q_-\|_{\mathrm{F}}^2 + \|Q_+\|_{\mathrm{F}}^2.$$

If $q > 0$, then $q^2 \ge 4q - 4$; summing over the at most b positive eigenvalues of Q gives

$$\|Q_+\|_{\mathrm{F}}^2 \ge 4\operatorname{tr} Q_+ - 4b. \tag{21}$$

Let $p_1 \ge \cdots \ge p_r \ge 0$ be the eigenvalues of $M = V^*V$ (counted with multiplicity, hence including any zeros). The nonzero eigenvalues of $P = VV^*$ are the same; extend the eigenvalue list of P by zeros as necessary. Likewise write $n_1 \ge n_2 \ge \cdots \ge 0$ for the eigenvalues of Q_- and extend either list by zeros to a common length. Von Neumann's trace inequality then implies

$$\operatorname{tr}(PQ_-) \le \sum_{i \le r} p_i n_i.$$

Consequently

$$\|P-Q_-\|_{\mathrm{F}}^2 + 4\operatorname{tr} Q_- \ge \sum_{i \le r} \left((p_i - n_i)^2 + 4n_i\right).$$

For fixed $p \ge 0$,

$$\min_{n \ge 0}\left((p-n)^2 + 4n\right) = \begin{cases} p^2, & 0 \le p \le 2, \\ 4p-4, & p \ge 2, \end{cases} = 2p - 1 + \Psi(p).$$

Therefore

$$\|P-Q_-\|_{\mathrm{F}}^2 \ge 2\operatorname{tr} P - r + \mathcal{D}(M) - 4\operatorname{tr} Q_-. \tag{22}$$

Combining (21) and (22),

$$\|P+Q\|_{\mathrm{F}}^2 \ge 4\operatorname{tr}(P+Q) - 2\operatorname{tr} P - r - 4b + \mathcal{D}(M).$$

Finally $\operatorname{tr} P$ is the sum of the squared column norms of V, so $\operatorname{tr} P \le r$, proving (20). □

Before stating the global consequence, define the retained central set

$$\mathcal{S}^\circ := \{\rho \in \mathcal{S}_1 : T < \gamma_\rho \leq 2T \text{ and } \rho \text{ lies outside the two boundary strips of Section 4}\}. \tag{23}$$

Let $S^\circ := \#\mathcal{S}^\circ$ and let

$$M^\circ := \big(\langle v_\rho, v_{\rho'}\rangle\big)_{\rho,\rho' \in \mathcal{S}^\circ}. \tag{24}$$

Thus M° is a principal submatrix of M and contains only simple critical-line zeros whose ordinates lie in $(T, 2T]$ and for which the compact kernel approximation is uniform.

Corollary 3.3 (Global Gram-defect inequality)**.** *Then*

$$N_0^s(T, 2T) \geq H_{\mathrm{MT}} N(T, 2T) + \mathcal{D}(M^\circ) - o(N(T, 2T)). \tag{25}$$

Proof. Apply Lemma 3.2 to $\widehat{A} = P_1 + Q'$ with $r = s_1$ and $b = s_2 + p$. By (14) and (15),

$$\begin{aligned} \|\widehat{A}\|_{\mathrm{F}}^2 &\geq 4 \operatorname{tr} \widehat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \\ &\geq 4 \operatorname{tr} \widehat{A} - s_1 - 2N(I') + \mathcal{D}(M). \end{aligned}$$

Hence

$$s_1 \geq 4 \operatorname{tr} \widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 - 2N(I') + \mathcal{D}(M). \tag{26}$$

Apply the independently proved tail estimate (16). Pinching M to the retained central principal block and its complement does not increase $\operatorname{tr}\Psi$: pinching is a random-unitary average, and the spectral trace functional $X \mapsto \operatorname{tr}\Psi(X)$ is convex and unitarily invariant. No operator convexity of Ψ is required. Thus $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$. Moreover,

$$N(I') = N(T, 2T) + o(N(T, 2T)).$$

Because s_1 counts simple critical-line zeros in the enlarged interval I', whereas $N_0^s(T, 2T)$ counts only those in I, we also use

$$s_1 \leq N_0^s(T, 2T) + N(I' \setminus I). \tag{27}$$

The standard local zero-count estimate gives

$$N(I' \setminus I) \ll D_0 \log T = T^{1/2} \log T = o(N(T, 2T)),$$

and hence

$$N_0^s(T, 2T) \geq s_1 - o(N(T, 2T)). \tag{28}$$

Substitution of (16), (17), and (18) into (26), followed by (28), gives (25). $\square$

4 The limiting overlap kernel

Put

$$h := \frac{2\pi}{L}, \qquad x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

Define

$$K(x) := \int_{-1/2}^{1/2} \cos(\sqrt{2}\, t) \cos(2\pi x t)\, dt \tag{29}$$

$$= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \tag{30}$$

$$k(x) := \frac{K(x)}{K(0)}, \qquad w(x) := k(x)^2, \qquad K(0) = \sqrt{2}\sin(1/\sqrt{2}). \tag{31}$$

The apparent singularities in (30) are removable.

Lemma 4.1 (Compact overlap limit)**.** *Fix $R_0 > 0$. Delete simple zeros lying in boundary strips of normalized width L^2 at the two ends of the sampling grid. The number deleted is $o(N(T, 2T))$. Uniformly for retained simple zeros satisfying $|x_\rho - x_{\rho'}| \le R_0$,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1). \tag{32}$$

Proof. By Appendix D, the independently proved Poisson–Gabor sampling identity, applied to the full grid, gives

$$\frac{1}{aL^2} \sum_{j \in \mathbb{Z}} \widehat{\phi}(\gamma - \tau_j) \widehat{\phi}(\gamma' - \tau_j) = \frac{\Phi(\gamma - \gamma')}{aL}, \qquad \Phi = \widehat{\phi^2}.$$

We record explicitly the pointwise truncation estimate needed here. The Fourier decay of the fixed-width tapered window gives, uniformly in L,

$$|\widehat{\phi}(r)| \ll \min\left\{L, \frac{1}{|r|}, \frac{1}{r^2}\right\}.$$

Suppose that both points lie at normalized distance at least L^2 from the finite grid ends and that $|x_\rho - x_{\rho'}| \le R_0$. If an omitted grid index lies $n \ge L^2$ grid steps beyond the nearer end, then

$$|\gamma - \tau_j| \asymp \frac{n}{L}, \qquad |\gamma' - \tau_j| \asymp \frac{n}{L}$$

uniformly for fixed R_0. Hence

$$|\widehat{\phi}(\gamma - \tau_j) \widehat{\phi}(\gamma' - \tau_j)| \ll \frac{L^4}{n^4}.$$

Summing the two omitted tails gives

$$\sum_{n \ge L^2} \frac{L^4}{n^4} \ll L^{-2}.$$

For the optimized window, dominated convergence also gives

$$a = \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 \, du = K(0) + o(1), \qquad K(0) = \sqrt{2} \sin(1/\sqrt{2}) > 0,$$

so in particular $a \asymp 1$. After multiplication by the Gram normalization $1/(aL^2)$, the omitted tails therefore contribute $O(L^{-4}) = o(1)$, uniformly for bounded normalized separation. Thus the full-grid Poisson–Gabor identity may be replaced by the finite sampled inner product with a uniform $o(1)$ error.

For $\gamma - \gamma' = hx$ and fixed x, substitute $u = Lt$:

$$\frac{\Phi(hx)}{aL} = \frac{\displaystyle\int_{-1/2}^{1/2} \phi(Lt)^2 e^{-2\pi i x t} \, dt}{\displaystyle\int_{-1/2}^{1/2} \phi(Lt)^2 \, dt}.$$

The squared tapered window converges in L^1 to $\cos(\sqrt{2}\, t) \mathbf{1}_{[-1/2, 1/2]}(t)$; the rescaled taper occupies length $O(L^{-1})$. Hence the last quotient converges to $K(x)/K(0)$, uniformly for x in compact sets. Combining this with the finite-grid tail bound proves (32).

Each deleted boundary strip has ordinate length $O(L)$, and the standard local zero-count estimate $N(t+1) - N(t) \ll \log(t+3)$ shows that only $O(L^2) = o(N)$ zeros are removed. ☐

Lemma 4.2 (Block defect)**.** *If $G \succeq 0$ is Hermitian, then*

$$\operatorname{tr}\Psi(G) \geq \min\left\{1,\ 2\sum_{i<j}|G_{ij}|^2\right\}. \tag{33}$$

Proof. If all eigenvalues of G lie in $[0,2]$, then $\Psi(G) = (G-I)^2$ and therefore

$$\operatorname{tr}\Psi(G) = \|G-I\|_{\mathrm{F}}^2 = \sum_i (G_{ii}-1)^2 + 2\sum_{i<j}|G_{ij}|^2 \geq 2\sum_{i<j}|G_{ij}|^2.$$

If some eigenvalue $\lambda > 2$, then its contribution is $\Psi(\lambda) = 2\lambda - 3 > 1$, while all other contributions are nonnegative. This proves (33). □

5 Position-weighted seven-point stability

Let $g_1, \dots, g_6 \geq 0$ be six consecutive gaps. For $1 \leq r \leq 6$ and $1 \leq i \leq 7-r$, write

$$s_{i,r} = g_i + \cdots + g_{i+r-1}$$

and define

$$\mathcal{E}_7(g) := \sum_{r=1}^{6} \frac{2}{7-r} \sum_{i=1}^{7-r} w(s_{i,r}). \tag{34}$$

For $p_j \geq 0$ put

$$\mathcal{F}_p(g) := \sum_{j=1}^{6} p_j g_j + \mathcal{E}_7(g). \tag{35}$$

Proposition 5.1 (Pressure redistribution)**.** *Let $\beta = \sum_{j=1}^{6} p_j$. If*

$$\mathcal{F}_p(g) \geq \delta \qquad (g \in \mathbb{R}^6_{\geq 0}),$$

then every ordered configuration $y_1 < \cdots < y_m$, $m \geq 7$, satisfies

$$E_m + \beta(y_m - y_1) \geq \delta(m-6), \qquad E_m := 2\sum_{1\leq i<j\leq m} w(y_j - y_i). \tag{36}$$

Proof. Sum the local inequality over the $m-6$ consecutive seven-point windows. A fixed pair separated by r gaps can appear in at most $7-r$ windows and has coefficient $2/(7-r)$ in each appearance; hence its total coefficient is at most 2. A fixed global gap can occupy each of the six relative gap positions at most once, so its accumulated pressure is at most $\sum_j p_j = \beta$. Therefore

$$(m-6)\delta \leq \sum_{\text{windows}} \mathcal{F}_p \leq E_m + \beta \sum_{j=1}^{m-1}(y_{j+1} - y_j),$$

which is (36). □

Proposition 5.2 (Computer-assisted seven-point inequality)**.** *For the pressure vector* (2), *every $g_1, \dots, g_6 \geq 0$ satisfies*

$$\frac{2714(g_1+g_6) + 3733(g_2+g_5) + 3553(g_3+g_4)}{10^7} + \sum_{r=1}^{6}\frac{2}{7-r}\sum_{i=1}^{7-r} w(g_i + \cdots + g_{i+r-1}) \geq \frac{39}{10000}. \tag{37}$$

Computer-assisted proof. The archived verifier performs exhaustive interval branch-and-bound with Arb/FLINT through `python-flint`. The pressure coefficients, target, and grid endpoints are represented by exact rationals. The verifier reconstructs rigorous Arb enclosures for w and its required derivatives from (30)–(31). The grid mesh is 1/4000.

Boxes are discarded only after one of three rigorous lower bounds proves the target: direct interval evaluation, the positive pressure bound, or a tangent bound after positive definiteness of the interval Hessian has been certified. Otherwise the widest coordinate is bisected. Reaching an uncertified terminal grid cell raises an exception; thus a successful run closes the entire search tree.

The frozen 256-bit run reports `verified=true`, with 1,119,372 visited nodes, 559,038 splits, and maximum depth 39. The same frozen verifier also succeeds at 128-bit working precision. Appendix E records the hashes and environment. □

Corollary 5.3. *For every $m \geq 7$ and $y_1 < \cdots < y_m$,*

$$E_m + \frac{1}{500}(y_m - y_1) \geq \frac{39}{10000}(m - 6). \tag{38}$$

Proof. Apply Proposition 5.1 to Proposition 5.2 and use $\sum_j p_j = 1/500$ from (2). □

6 Block stability

Take

$$m = 262, \qquad A_0 := \frac{39}{10000}(m - 6) = \frac{624}{625} < 1. \tag{39}$$

This is the largest integer m for which $(39/10000)(m - 6) < 1$.

Let B be a block of m consecutive retained simple zeros, let $\operatorname{span}(B)$ denote the difference between its extreme normalized ordinates, and let G_B be its Gram matrix.

Lemma 6.1 (Uniform block stability)**.** *Uniformly over such blocks,*

$$\mathcal{D}(G_B) + \frac{1}{500}\operatorname{span}(B) \geq A_0 - o(1). \tag{40}$$

Proof. If $\operatorname{span}(B)/500 \geq A_0$, the claim is immediate because $\mathcal{D}(G_B) \geq 0$. Otherwise $\operatorname{span}(B) < 500A_0 < 500$. Since m is fixed, all $\binom{m}{2}$ pair separations lie in one fixed compact interval. By Lemma 4.1,

$$2\sum_{i<j} |(G_B)_{ij}|^2 = E_m + o(1)$$

uniformly over the blocks in this branch. Lemma 4.2 gives

$$\mathcal{D}(G_B) \geq \min\{1, E_m + o(1)\}.$$

Corollary 5.3 gives

$$E_m \geq A_0 - \frac{1}{500}\operatorname{span}(B).$$

The right-hand side lies in $(0, A_0]$ and $A_0 < 1$. Hence

$$\mathcal{D}(G_B) \geq A_0 - \frac{1}{500}\operatorname{span}(B) - o(1),$$

which proves (40). Uniformity of the error follows from the fixed block size and the compact separation range. □

7 Shifted block pinching

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalized ordinates of the set $\mathcal{S}^\circ$ defined in (23). By the boundary estimate in Lemma 4.1,

$$S^\circ = N_0^s(T, 2T) - o(N(T, 2T)). \tag{41}$$

For each of the m offsets modulo m, partition the ordered points into full consecutive blocks of size m, leaving only $O(m)$ endpoint points. Pinching M° to the full principal blocks and the remainder yields

$$\mathcal{D}(M^\circ) \geq \sum_B \mathcal{D}(G_B). \tag{42}$$

Indeed, if $\mathcal{P}$ denotes this pinching, then for the finite group $\mathcal{U}$ of blockwise sign unitaries one has the random-unitary representation

$$\mathcal{P}(X) = \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} UXU^*.$$

Since the spectral trace functional $F(X) := \operatorname{tr} \Psi(X)$ is convex and unitarily invariant,

$$F(\mathcal{P}(X)) \leq \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} F(UXU^*) = F(X).$$

The pinched matrix is the direct sum of the full block Gram matrices and a remainder block. Since $\Psi \geq 0$, the latter has nonnegative defect and may be discarded, proving (42). No operator convexity of Ψ is used.

For a fixed offset, sum (40) over its full blocks; then average over all m offsets. The average number of full blocks is $S^\circ/m + O(1)$. Each consecutive gap $x_{j+1} - x_j$ belongs to a block span for at most $m - 1$ offsets. Moreover,

$$x_{S^\circ} - x_1 \leq \frac{T}{h} = \frac{LT}{2\pi} = N(T, 2T) + o(N(T, 2T)) \tag{43}$$

by the Riemann–von Mangoldt formula. Since the error in Lemma 6.1 is uniform and m is fixed,

$$\mathcal{D}(M^\circ) \geq \frac{A_0}{m} N_0^s(T, 2T) - \frac{m-1}{500m} N(T, 2T) - o(N(T, 2T)). \tag{44}$$

For (39),

$$\mathcal{D}(M^\circ) \geq \frac{312}{81875} N_0^s(T, 2T) - \frac{261}{131000} N(T, 2T) - o(N(T, 2T)). \tag{45}$$

Proof of Theorem 1.1. Insert (45) into Corollary 3.3:

$$N_0^s \geq H_{\mathrm{MT}} N + \frac{312}{81875} N_0^s - \frac{261}{131000} N - o(N).$$

Therefore

$$\frac{81563}{81875} N_0^s \geq \left(H_{\mathrm{MT}} - \frac{261}{131000} \right) N - o(N).$$

Divide by $N = N(T, 2T)$ and pass to the lower limit:

$$\begin{aligned}\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} &\geq \frac{H_{\mathrm{MT}} - 261/131000}{81563/81875} \\ &= \frac{655000 H_{\mathrm{MT}} - 1305}{652504} \\ &= 0.6730732086087052768351\ldots.\end{aligned}$$

□

8 Comparison and structural interpretation

Jo's upstream reproducible draft [13] uses $p_j = 1/3000$ and certifies $19/5000 = 0.0038$, leading to $0.6730085279277797\ldots$. Rademacher's subsequent reproducible refinement [14] keeps the same uniform pressure and strengthens the certified local value to $191/50000 = 0.00382$, leading to $0.6730213619501665\ldots$. Our vector has the same total pressure $1/500$ but certifies $39/10000 = 0.0039$.

The natural finite-dimensional optimization exposed by Proposition 5.1 is

$$\sup_{\substack{p_j \geq 0 \\ \sum p_j = 1/500}} \inf_{g \in \mathbb{R}^6_{\geq 0}} \left(\sum_{j=1}^{6} p_j g_j + \mathcal{E}_7(g) \right). \tag{46}$$

The uniform vector is one feasible point of this problem. The strongest reproduced *certificate* for that uniform vector among the two artifacts just cited is Rademacher's $191/50000 = 0.00382$, whereas Proposition 5.2 gives a certified value $39/10000 = 0.0039$ for the position-weighted vector (2). This comparison proves that the new *certified lower bound* is stronger; by itself it does not prove that the exact local infimum for the uniform vector is below 0.0039. Accordingly, we make no claim here that uniform pressure is globally suboptimal, nor that (2) solves (46).

A larger optimization could also redistribute pair-energy coefficients $c_{r,i}$ subject to the double-counting budgets $\sum_{i=1}^{7-r} c_{r,i} \leq 2$. We leave that problem open.

9 Scope and trust base

Logical status of the theorem. The common compression is fixed in Section 2, and the analytic/Gabor package used in the main argument is proved in Appendices A–D; it is not imported from Theorem D of [5]. The analytic external trust base consists of standard classical results: the Weil explicit formula for ζ in the Fourier normalization stated in Appendix C, Stirling's formula, the Prime Number Theorem, the Riemann–von Mangoldt formula and its standard local zero-count consequence, standard Chebyshev and prime-power estimates, the weighted Montgomery–Vaughan Hilbert inequality, and Poisson summation. The Montgomery–Vaughan paper is correctly cited as *Hilbert's inequality*, J. London Math. Soc. (2) **8** (1974), 73–82.

The role of [5] in this version is comparison and normalization cross-checking only. No proposition from that preprint is invoked as a premise of Theorem 1.1. Likewise, the recent pair-correlation papers cited in the introduction provide context; the particular first/second-moment route used here is reconstructed directly in Appendix C.

Proposition 5.2 is computer-assisted. Its trust base includes the frozen verifier, `python-flint`, FLINT/Arb and their dependencies, and the correctness of the execution environment. The 128-

and 256-bit runs are replications of the same implementation at different working precisions, not algorithmically independent verifiers.

The pressure vector (2) is an explicit certified improvement, not an optimality theorem. Thus the paper is self-contained relative to the classical analytic results just listed and to the stated Arb/FLINT certificate; it does not claim to reprove those classical theorems from first principles.

A Independent proof of Input I: zero-side inertia

This appendix proves the algebraic finite-compression statement used in Proposition 3.1. The horizontal partner $\rho^\sharp = 1 - \overline{\rho}$ is the composition of conjugation with the functional-equation symmetry; it preserves ordinary ordinate.

A.1 Scope

Let I' be a finite ordinate interval. Since $\rho \mapsto 1 - \overline{\rho}$ preserves ordinary ordinate, the zeros with ordinates in I' form a pairing-invariant finite multiset $Z(I')$ of nontrivial zeta zeros whose ordinates lie in I'. We partition the *distinct* zeros into

$$\mathcal{S}_1 = \{\text{simple critical-line points}\}, \qquad \mathcal{S}_2 = \{\text{multiple critical-line points}\},$$

and a set $\mathcal{P}$ containing one representative from each off-line pair

$$\{\rho, \rho^\sharp\}, \qquad \rho^\sharp := 1 - \overline{\rho}.$$

Write

$$s_1 = \#\mathcal{S}_1, \qquad s_2 = \#\mathcal{S}_2, \qquad p = \#\mathcal{P}.$$

The only zeta-specific fact used in the inertia calculation is the combined functional-equation/conjugation symmetry, which preserves multiplicity:

$$m_{\rho^\sharp} = m_\rho.$$

The finite zero-side Hermitian form obtained by compressing Weil's form to a finite test family can then be written as a pull-back of the block form described below. The present note verifies the resulting inertia statement from scratch.

A.2 Positive index under pull-back

For a Hermitian form H on a finite-dimensional complex vector space, let $n_+(H)$ denote its positive index: the maximal dimension of a subspace on which H is positive definite.

Lemma A.1 (Pull-back monotonicity)**.** *Let H be a Hermitian form on $\mathbb{C}^m$, and let $A : U \to \mathbb{C}^m$ be either complex-linear or conjugate-linear. Then*

$$n_+(H \circ A) \le n_+(H).$$

Proof. Let $U_0 \subset U$ be a complex subspace on which $H \circ A$ is positive definite. Then $A|_{U_0}$ is injective. If A is conjugate-linear, $A(U_0)$ is still a complex subspace, because $\alpha A(u) = A(\overline{\alpha}\, u)$ for every $\alpha \in \mathbb{C}$. Moreover H is positive definite on $A(U_0)$. Hence

$$\dim_{\mathbb{C}} U_0 = \dim_{\mathbb{C}} A(U_0) \le n_+(H).$$

Taking the maximum over U_0 proves the claim. $\square$

Corollary A.2 (Subadditivity)**.** *For Hermitian forms H_1, H_2 on the same space,*

$$n_+(H_1 + H_2) \le n_+(H_1) + n_+(H_2).$$

Proof. Apply Lemma A.1 to $H_1 \oplus H_2$ under the diagonal map $u \mapsto (u, u)$. □

A.3 The off-line hyperbolic block

Lemma A.3 (Signature of one horizontal symmetry pair)**.** *Let $m > 0$. On $\mathbb{C}^2$ define*

$$H_m(x, y) = 2m \operatorname{Re}(x\overline{y}).$$

Then

$$H_m(x, y) = \begin{pmatrix} \overline{x} & \overline{y} \end{pmatrix} \begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

and the form has signature $(1, 1)$.

Proof. The Hermitian matrix

$$B_m = \begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}$$

has characteristic polynomial $\lambda^2 - m^2$, hence eigenvalues m and $-m$. Equivalently, with

$$u = \frac{x + y}{\sqrt{2}}, \qquad v = \frac{x - y}{\sqrt{2}},$$

one has

$$H_m(x, y) = m|u|^2 - m|v|^2.$$

Thus $n_+(H_m) = n_-(H_m) = 1$. □

This is the entire source of the "one positive direction per off-line pair" rule. No limiting argument and no assumption concerning the distance of the pair from the critical line is involved.

A.4 The zero-side block form

For each distinct critical-line zero ρ, let m_ρ be its multiplicity. Introduce one evaluation coordinate $z_\rho \in \mathbb{C}$. For each off-line representative $\rho \in \mathcal{P}$, introduce two coordinates $(z_\rho, z_{\rho^\sharp})$.

Define the Hermitian form

$$\begin{aligned} \mathcal{Q}(z) = &\sum_{\rho \in \mathcal{S}_1 \cup \mathcal{S}_2} m_\rho |z_\rho|^2 \\ &+ \sum_{\rho \in \mathcal{P}} 2m_\rho \operatorname{Re}\big(z_\rho \overline{z_{\rho^\sharp}}\big). \end{aligned} \tag{47}$$

Proposition A.4 (Exact block inertia)**.** *The form $\mathcal{Q}$ is an orthogonal direct sum of $s_1 + s_2$ positive one-dimensional blocks and p hyperbolic two-dimensional blocks. Consequently*

$$n_+(\mathcal{Q}) = s_1 + s_2 + p.$$

If the simple critical-line blocks are removed, the remaining form $\mathcal{Q}'$ satisfies

$$n_+(\mathcal{Q}') = s_2 + p.$$

Proof. Every critical-line block is $m_\rho |z|^2$ with $m_\rho > 0$, and hence has positive index one. Every off-line pair block has signature $(1, 1)$ by Lemma A.3. Positive index is additive under orthogonal direct sum, proving both assertions. □

A.5 Compression by evaluation

Let $U = \mathbb{C}^d$ be the coefficient space of the finite family of test functions used to form the compression. For each zero parameter appearing on the zero side, let ℓ_ρ denote the conjugate-linear coefficient functional $c \mapsto \Lambda_\rho(c)$ from Section 2. Define

$$\mathfrak{E}(c) := \big(\ell_\rho(c)\big)_\rho.$$

The map $\mathfrak{E}$ is conjugate-linear in the displayed coefficient convention; Lemma A.1 was formulated to cover this case directly.

By Proposition 2.1, the normalized finite zero-side matrix $\widehat{A}$ is exactly the matrix of the pull-back form

$$c \longmapsto \mathcal{Q}(\mathfrak{E}c).$$

Thus this inertia calculation applies to the same matrix $\widehat{A}$ used in the main argument.

For a simple critical-line zero $\rho \in \mathcal{S}_1$, use the normalized evaluation vector $v_\rho \in \mathbb{C}^d$ defined in Proposition 2.1; thus the normalized evaluation is $c^* v_\rho$. Proposition 2.1 proves for this same family that

$$\|v_\rho\| \le 1.$$

Set

$$P_1 = \sum_{\rho \in \mathcal{S}_1} v_\rho v_\rho^*, \qquad M = V^* V,$$

where V has the v_ρ as columns, and put

$$Q' = \widehat{A} - P_1.$$

Theorem A.5 (Independent Input I)**.** *With the preceding notation,*

$$P_1 \succeq 0, \qquad \operatorname{rank} P_1 \le s_1, \qquad \operatorname{tr} P_1 \le s_1,$$

and

$$n_+(Q') \le s_2 + p.$$

In particular, the finite compression cannot acquire more than one positive direction from each multiple critical-line point and each off-line horizontal symmetry pair after the simple critical-line contribution has been separated.

Proof. Each summand $v_\rho v_\rho^*$ is positive semidefinite and rank at most one, so

$$P_1 \succeq 0, \qquad \operatorname{rank} P_1 \le s_1.$$

Moreover

$$\operatorname{tr} P_1 = \sum_{\rho \in \mathcal{S}_1} \|v_\rho\|^2 \le s_1.$$

After deleting the simple critical-line coordinates from the block form $\mathcal{Q}$, the remaining form is $\mathcal{Q}'$ of Proposition A.4, with

$$n_+(\mathcal{Q}') = s_2 + p.$$

The matrix Q' on coefficient space is the pull-back of this form under the corresponding restricted evaluation map. Lemma A.1 therefore gives

$$n_+(Q') \le n_+(\mathcal{Q}') = s_2 + p.$$

□

A.6 Why the functional-equation/conjugation pairing produces the block

For completeness we isolate the algebra behind (47). Weil's Hermitian zero-side form pairs the evaluation at ρ with the evaluation at the horizontal partner under functional-equation plus conjugation symmetry

$$\rho^\sharp = 1 - \overline{\rho}.$$

For an off-line pair of equal multiplicity m_ρ, its two contributions combine as

$$m_\rho \left(\ell_\rho(c)\overline{\ell_{\rho^\sharp}(c)} + \ell_{\rho^\sharp}(c)\overline{\ell_\rho(c)} \right) = 2m_\rho \operatorname{Re} \left(\ell_\rho(c)\overline{\ell_{\rho^\sharp}(c)} \right),$$

which is exactly the hyperbolic block of Lemma A.3. On the critical line, $\rho = \rho^\sharp$, so the same expression collapses to a positive square (up to the normalization convention already absorbed into the zero-side form).

Thus the signature claim is a direct algebraic consequence of functional equation symmetry once the zero-side Hermitian form has been written in evaluation coordinates.

A.7 Adversarial checks

Several possible pathologies do not change the result.

- **Near-line pairs.** As $\operatorname{Re} \rho \to 1/2$, the two evaluation functionals may become nearly dependent, but pull-back can only *decrease* positive index. No uniform linear independence is required.
- **Deep off-line pairs.** The eigenvalues $\pm m_\rho$ of the abstract pair block are independent of the depth $|\operatorname{Re} \rho - 1/2|$. The evaluation map may distort or annihilate directions, but cannot increase n_+.
- **Coincident ordinates.** Different zero points may have the same ordinate. This creates dependence among evaluation functionals, which again can only lower the pull-back positive index.
- **Multiplicity.** A multiple critical-line point contributes one positive coordinate block $m_\rho|z|^2$, hence one positive direction, regardless of m_ρ. Multiplicity changes the weight but not its positive index.
- **Noninjective compression.** No injectivity of the full evaluation map is assumed; this is precisely why Lemma A.1 is formulated for arbitrary linear maps.

B Independent proof of Input II: exterior-zero tail

This appendix proves the trace-class tail estimate and its effect on the rank–trace functional. The exterior matrix is grouped into Hermitian critical-line squares and off-line pair terms $uv^* + vu^*$, so that $G = A + E$ holds entrywise under the same zero-side convention.

B.1 Setup

Let

$$\ell = \log \frac{T}{2\pi}, \qquad L = \ell, \qquad X = e^L = \frac{T}{2\pi}, \qquad D_0 = T^{1/2},$$

and let

$$I = (T, 2T], \qquad I' = (T - D_0,\, 2T + D_0].$$

Let the critical sampling grid be

$$\tau_k = T + \frac{2\pi k}{L}, \qquad 0 \le k < d, \qquad d = \left\lfloor \frac{LT}{2\pi} \right\rfloor.$$

The tapered Montgomery–Taylor window ϕ has support in $[-L/2, L/2]$, satisfies $0 \le \phi \le 1$, and for fixed taper width has

$$C_1 := \|\phi''\|_1 = O(1).$$

Put

$$a = \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 \, du.$$

For this window, $a \to K(0) > 0$; hence $a \asymp 1$.

Use the globally fixed test family (4). Thus

$$G_{jk} = W(f_j, f_k)$$

is exactly the common matrix (8). Using the zero-side expansion of the normalized Weil formula, define A_{jk} by restricting that zero sum to zeros whose ordinary ordinates lie in I', grouping every off-line zero together with its horizontal partner $\rho^\sharp = 1 - \overline{\rho}$. Define E_{jk} by the complementary exterior zero sum. Since the inside and outside sets form a disjoint partition invariant under $\rho \mapsto \rho^\sharp$, one has entrywise

$$\boxed{G = A + E.}$$

Equivalently,

$$\widehat{A} = \widehat{G} - \widehat{E}, \qquad \widehat{G} = \frac{G}{aL^2}, \quad \widehat{A} = \frac{A}{aL^2}, \quad \widehat{E} = \frac{E}{aL^2}.$$

B.2 Off-axis Fourier decay

We use the standard Paley–Wiener integration-by-parts bound.

Lemma B.1 (Complex-shift decay)**.** *Let $\phi \in C_c^2([-L/2, L/2])$. For real $r \neq 0$ and $|y| < 1/2$,*

$$|\widehat{\phi}(r - iy)| \le e^{L|y|/2} \, \|\phi''\|_1 \, |r - iy|^{-2} \le X^{1/4} C_1 \, |r|^{-2}.$$

Proof. Twice integrating by parts in

$$\widehat{\phi}(r - iy) = \int_{-L/2}^{L/2} \phi(u) e^{-iru - yu} \, du$$

and using the compactly supported C^2 taper gives

$$\widehat{\phi}(r - iy) = -(r - iy)^{-2} \int \phi''(u) e^{-i(r - iy)u} \, du.$$

Since $|e^{-yu}| \le e^{L|y|/2}$ on the support,

$$|\widehat{\phi}(r - iy)| \le e^{L|y|/2} C_1 |r - iy|^{-2}.$$

Now $|r - iy| \ge |r|$ and, because $|y| < 1/2$,

$$e^{L|y|/2} \le e^{L/4} = X^{1/4}.$$

$\square$

B.3 One external zero

Let $\rho = \beta + i\gamma$ be a distinct zero with $\gamma \notin I'$ and put

$$y = \beta - \frac{1}{2}, \qquad |y| < \frac{1}{2}.$$

Define its sampled vector

$$u_\rho = \left(\widehat{\phi}((\gamma - \tau_k) - iy)\right)_{0 \le k < d}.$$

Let

$$D = \mathrm{dist}(\gamma, I) \ge D_0.$$

Lemma B.2 (Tail vector bound)**.** *Uniformly for every such zero,*

$$\|u_\rho\|_2^2 \ll C_1^2 X^{1/2} L D^{-3}.$$

Proof. Lemma B.1 gives

$$\|u_\rho\|_2^2 \le C_1^2 X^{1/2} \sum_{0 \le k < d} |\gamma - \tau_k|^{-4}.$$

The grid spacing is $h = 2\pi/L$. Comparing the monotone sum with an integral,

$$\sum_{0 \le k < d} |\gamma - \tau_k|^{-4} \ll D^{-4} + \frac{1}{h}\int_D^\infty t^{-4}\,dt \ll D^{-4} + LD^{-3} \ll LD^{-3},$$

because $D \ge 1$ and $L \ge 2$ for large T. □

B.4 Summing over zeros outside the enlarged interval

We use only the classical estimate

$$N(t+1) - N(t) \le A_0 \log(t+3) \tag{48}$$

for some absolute A_0 and all $t \ge 0$.

Lemma B.3 (Weighted exterior zero count)**.** *Counting zeros with multiplicity,*

$$\sum_{\gamma \notin I'} m_\rho \ \mathrm{dist}(\gamma, I)^{-3} \ll \frac{\log T}{D_0^2}.$$

Proof. For the upper tail, group zeros into intervals

$$2T + D_0 + j < \gamma \le 2T + D_0 + j + 1, \qquad j \ge 0.$$

By (48), the contribution is

$$\ll \sum_{j \ge 0} \frac{\log(2T + D_0 + j + 4)}{(D_0 + j)^3}.$$

Splitting at $j = T$ gives

$$\ll \frac{\log T}{D_0^2}.$$

The lower positive tail $0 < \gamma < T - D_0$ is identical. Zeros with $\gamma \le 0$ are at distance $\gg T$ from I and contribute $O(T^{-2}\log T)$, which is smaller. Adding the three ranges proves the claim. □

B.5 Trace-norm tail bound

The exterior zero-side matrix must be grouped according to the horizontal symmetry involution. If u_ρ is the sampled evaluation column, then

$$E = \sum_{\substack{\rho \text{ exterior} \\ \rho=\rho^\sharp}} m_\rho u_\rho u_\rho^* + \sum_{\substack{\{\rho,\rho^\sharp\} \text{ exterior} \\ \rho\neq\rho^\sharp}} m_\rho \left(u_\rho u_{\rho^\sharp}^* + u_{\rho^\sharp} u_\rho^* \right).$$

This is Hermitian term-by-term after pairing. Moreover

$$\|uv^* + vu^*\|_1 \le 2\|u\|_2\|v\|_2 \le \|u\|_2^2 + \|v\|_2^2.$$

Hence the scalar tail-vector estimate may be summed exactly as before.

Proposition B.4 (Independent tail estimate)**.** *One has*

$$\|E\|_1 \ll C_1^2 X^{1/2} L \frac{\log T}{D_0^2}.$$

Consequently

$$\|\widehat{E}\|_1 = \frac{\|E\|_1}{aL^2} \ll \frac{X^{1/2}\log T}{LD_0^2}.$$

At $\lambda = 1$, $L = \ell$, $X = T/(2\pi)$ and $D_0 = T^{1/2}$, hence

$$\boxed{\|\widehat{E}\|_1 \ll T^{-1/2}}.$$

Proof. For critical-line exterior zeros use $\|u_\rho u_\rho^*\|_1 = \|u_\rho\|_2^2$. For each off-line pair use the preceding inequality. Both members have the same ordinate and opposite horizontal displacement, so Lemma B.2 applies to each. Summing with multiplicity and using Lemma B.3 gives the stated unnormalised bound. Division by aL^2, with $a \asymp 1$, gives the normalized estimate. ☐

B.6 The rank–trace functional under a trace-class perturbation

Define

$$\mathcal{R}(H) = 4\operatorname{tr} H - \|H\|_{\mathrm{F}}^2$$

for a finite Hermitian matrix H.

Lemma B.5 (Perturbation inequality)**.** *If $A = G - E$ are Hermitian matrices and $\varepsilon = \|E\|_1$, then*

$$\mathcal{R}(A) \ge \mathcal{R}(G) - 4\varepsilon - 2\|G\|_{\mathrm{F}}\varepsilon - \varepsilon^2. \tag{49}$$

Proof. Since $|\operatorname{tr} E| \le \|E\|_1 = \varepsilon$,

$$4\operatorname{tr} A \ge 4\operatorname{tr} G - 4\varepsilon.$$

Also

$$\|A\|_{\mathrm{F}} \le \|G\|_{\mathrm{F}} + \|E\|_{\mathrm{F}} \le \|G\|_{\mathrm{F}} + \|E\|_1 = \|G\|_{\mathrm{F}} + \varepsilon.$$

Squaring and combining the two inequalities proves (49). ☐

Theorem B.6 (Independent Input II)**.** *For the optimized bandwidth $\lambda = 1$, the prime-side moment estimate of Appendix C gives $\|\widehat{G}\|_{\mathrm{F}} = O(N(T,2T)^{1/2})$. Then*

$$\boxed{4\operatorname{tr}\widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 \ge 4\operatorname{tr}\widehat{G} - \|\widehat{G}\|_{\mathrm{F}}^2 - o(N(T,2T)).}$$

Proof. Apply Lemma B.5 to the normalized matrices with

$$\varepsilon = \|\widehat{E}\|_1 \ll T^{-1/2}$$

from Proposition B.4. Since $N(T, 2T) \asymp T \log T$,

$$\varepsilon = o(N), \qquad \varepsilon^2 = o(N),$$

and using the Appendix C bound,

$$\|\widehat{G}\|_{\mathrm{F}}\varepsilon = O(N^{1/2}T^{-1/2}) = o(N).$$

Substitution in (49) proves the result. ☐

Remark B.7 (Package closure)**.** *Appendix C establishes the prime-side second moment without using the present tail comparison. Thus using $\|\widehat{G}\|_{\mathrm{F}} = O(N^{1/2})$ here is logically acyclic.*

C Independent proof of Input III: prime-side moments

This appendix fixes the Fourier normalization of Weil's explicit formula [1, 4], derives the spectral density, controls integrated finite-grid end effects, and evaluates the prime diagonal and both off-diagonal frequency types. The weighted Montgomery–Vaughan inequality [3] is used only with an unspecified absolute constant.

C.1 Classical input and notation

Put

$$\ell = \log \frac{T}{2\pi}, \qquad L = \ell, \qquad X = e^L = \frac{T}{2\pi}, \qquad I = [T, 2T],$$

and define

$$\ell_1 := \ell + 2\log 2 - 1. \tag{50}$$

Thus the Riemann–von Mangoldt formula reads

$$N(T, 2T) = \frac{T\ell_1}{2\pi} + O(\ell).$$

Let $\phi = \phi_L \in C_c^2(\mathbb{R})$ be real, even, supported exactly on $[-L/2, L/2]$, with a fixed-width endpoint taper and uniform derivative bounds. Define

$$\widehat{\phi}(r) = \int_{\mathbb{R}} \phi(u)e^{-iru}\,du, \qquad \Phi(r) = \int_{\mathbb{R}} \phi(u)^2 e^{-iru}\,du,$$

$$a = \frac{1}{L}\int \phi^2, \qquad b = \frac{1}{L}\int \phi^4, \qquad g = \phi^2 * \phi^2.$$

For the optimized family, a and b remain bounded above and below by positive constants.

We use the classical Weil explicit formula in the following fixed normalization. With

$$\widehat{q}(z) = \int_{\mathbb{R}} q(u)e^{izu}\,du, \qquad q(u) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{q}(t)e^{-itu}\,dt,$$

the external theorem used here is

$$\sum_\rho m_\rho \widehat{q}(z_\rho) = \widehat{q}(i/2) + \widehat{q}(-i/2) - \sum_{n\ge 1} \frac{\Lambda(n)}{\sqrt{n}} (q(\log n) + q(-\log n)) \\ + \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{q}(\tau) \left[\Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{i\tau}{2} \right) - \log \pi \right] d\tau. \tag{51}$$

Here $z_\rho = -i(\rho - \frac{1}{2})$ and zeros are counted with multiplicity. This is the standard Guinand–Weil explicit formula in the present Fourier normalization; see Weil [1] and, for a modern formulation with explicit strip hypotheses, Iwaniec–Kowalski [2, Theorem 5.12]. We apply it to $q = f * \widetilde{g}$, $\widetilde{g}(u) = \overline{g(-u)}$. Since $f, g \in C_c^2([-L/2, L/2])$, one has $q \in C_c^2([-L, L])$. Moreover

$$\widehat{q}(z) = \mathcal{F}_+ f(z) \, \overline{\mathcal{F}_+ g(\overline{z})},$$

and two integrations by parts in each factor give, uniformly on every fixed strip $|\Im z| \le \frac{1}{2} + \varepsilon$,

$$|\widehat{q}(z)| \ll_{\varepsilon, L, f, g} (1 + |z|)^{-4}.$$

Thus the standard strip-decay hypotheses of the explicit formula are satisfied. The support condition also implies that only $n \le X = e^L$ occur in the prime sum. Fourier inversion then gives the following proposition.

Proposition C.1 (Normalized Weil spectral density)**.** *The compressed Weil form has density*

$$\nu_X(\tau) = \mu(\tau) + \Pi_X(\tau) + P_X(\tau),$$

where

$$\mu(\tau) = \frac{1}{2\pi} \Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{i\tau}{2} \right) - \frac{\log \pi}{2\pi}, \tag{52}$$

$$P_X(\tau) = -\frac{1}{\pi} \sum_{n \le X} \frac{\Lambda(n)}{\sqrt{n}} \cos(\tau \log n), \tag{53}$$

$$\Pi_X(\tau) = \frac{1}{2\pi(1/4 + \tau^2)} + \frac{1}{\pi} \Re \frac{X^{1/2 + i\tau} - 1}{1/2 + i\tau}. \tag{54}$$

More precisely,

$$W(f, g) = \int_{\mathbb{R}} \mathcal{F}_+ f(\tau) \overline{\mathcal{F}_+ g(\tau)} \nu_X(\tau) \, d\tau.$$

Proof. The gamma contribution in the classical explicit formula is the integral against μ. For the prime term,

$$q(y) + q(-y) = \frac{1}{\pi} \int_{\mathbb{R}} \mathcal{F}_+ f(\tau) \overline{\mathcal{F}_+ g(\tau)} \cos(\tau y) \, d\tau,$$

and $q(\pm \log n) = 0$ for $n > X$. For the pole terms,

$$H(i/2) + H(-i/2) = \int_{\mathbb{R}} q(u)(e^{-u/2} + e^{u/2}) \, du.$$

Since $\operatorname{supp} q \subset [-L, L]$, this equals

$$\int_{\mathbb{R}} q(u) e^{-|u|/2} \, du + \int_{-L}^{L} q(u) e^{|u|/2} \, du.$$

With $s = 1/2 + i\tau$,
$$\int_{\mathbb{R}} e^{-|u|/2} e^{-i\tau u} du = (1/4 + \tau^2)^{-1},$$
while
$$\int_{-L}^{L} e^{|u|/2} e^{-i\tau u} du = 2\Re \frac{X^{\bar{s}} - 1}{\bar{s}} = 2\Re \frac{X^s - 1}{s}.$$
Fourier inversion therefore gives exactly the density Π_X. This proves the spectral identity with all signs and 2π factors fixed. □

Stirling gives, uniformly for $\tau \in I$,
$$\mu(\tau) = \frac{1}{2\pi} \log \frac{\tau}{2\pi} + O(T^{-2}), \qquad \mu'(\tau) = O(T^{-1}).$$
Also
$$|P_X(\tau)| \ll \sqrt{X}, \qquad |\Pi_X(\tau)| \ll \frac{\sqrt{X}}{1 + |\tau|},$$
and Riemann–von Mangoldt yields
$$\int_T^{2T} \mu(\tau) d\tau = N(T, 2T) + O(\ell).$$

Let
$$h = \frac{2\pi}{L}, \qquad d = \left\lfloor \frac{T}{h} \right\rfloor, \qquad \tau_k = T + kh \quad (0 \le k < d).$$
By (9), the same common matrix G satisfies
$$G_{jk} = \int_{\mathbb{R}} \widehat{\phi}(\tau - \tau_j) \widehat{\phi}(\tau - \tau_k) \nu_X(\tau)\, d\tau,$$
and set
$$\widetilde{G} := \frac{G}{L}, \qquad \widehat{G} := \frac{\widetilde{G}}{aL} = \frac{G}{aL^2}.$$
This distinction between G, $\widetilde{G}$ and $\widehat{G}$ is maintained throughout.

C.2 Fourier and prime-power estimates

The fixed-width C^2 taper gives
$$\max\{|\widehat{\phi}(r)|, |\Phi(r)|\} \le \psi(r) := \min\left\{L, \frac{C_1}{|r|}, \frac{C_2}{r^2}\right\}, \tag{55}$$
with constants independent of L. Consequently
$$\Psi_0 := \int_0^\infty \psi(r)\, dr \ll \log L, \qquad \int_{\mathbb{R}} \psi(r)^2 |r|\, dr \ll \log L, \qquad \int_{\mathbb{R}} \psi(r)^2\, dr \ll L. \tag{56}$$
Plancherel and Fourier inversion give
$$\int_{\mathbb{R}} \Phi(x)^2\, dx = 2\pi b L, \qquad \int_{\mathbb{R}} \Phi(x)^2 e^{ixy}\, dx = 2\pi g(y). \tag{57}$$

We also use the classical prime-power estimates

$$\sum_{n\le x}\frac{\Lambda(n)}{\sqrt{n}}\ll\sqrt{x},\qquad \sum_{n\le x}\Lambda(n)^2\ll x\log x, \tag{58}$$

and

$$\sum_{n\le x}\frac{\Lambda(n)^2}{n}=\frac{1}{2}\log^2 x+O(\log x). \tag{59}$$

The asymptotic (59) follows from the Prime Number Theorem by standard partial summation; the upper bounds in (58) require only Chebyshev-strength estimates. We use Titchmarsh [4] as a standard reference.

C.3 First trace moment

Put

$$S_d(\tau):=\sum_{0\le k<d}|\widehat{\phi}(\tau-\tau_k)|^2,\qquad S_\infty(\tau):=\sum_{k\in\mathbb{Z}}|\widehat{\phi}(\tau-\tau_k)|^2.$$

For the infinite translated critical grid, Poisson summation gives

$$S_\infty(\tau)=aL^2\qquad(\tau\in\mathbb{R}). \tag{60}$$

We require two distinct end-effect estimates. The first controls the lattice points omitted from the finite grid while τ remains in I:

$$R_{\mathrm{in}}:=\int_I(S_\infty(\tau)-S_d(\tau))|\nu_X(\tau)|\,d\tau.$$

On I one has

$$|\nu_X(\tau)|\ll B,\qquad B:=\ell+\sqrt{X},$$

by Stirling and the displayed bounds for P_X,Π_X. Using (55), the left omitted tail is bounded by

$$\sum_{j\ge1}\int_{jh}^{\infty}\psi(r)^2\,dr,$$

and the right omitted tail is identical up to $O(1)$ endpoint indices. Hence, by Tonelli and (56),

$$R_{\mathrm{in}}\ll B\left(\frac{1}{h}\int_0^\infty r\psi(r)^2\,dr+\int_0^\infty\psi(r)^2\,dr\right)\ll BL\log L. \tag{61}$$

The second estimate, absent from a naive truncation to I, controls the actual finite-grid mass outside I:

$$R_{\mathrm{out}}:=\int_{\mathbb{R}\setminus I}S_d(\tau)|\nu_X(\tau)|\,d\tau.$$

For the prime and pole pieces the global bounds

$$|P_X(\tau)|\ll\sqrt{X},\qquad|\Pi_X(\tau)|\ll1+\sqrt{X}$$

give, by the same endpoint-tail summation,

$$\int_{\mathbb{R}\setminus I}S_d(\tau)(|P_X(\tau)|+|\Pi_X(\tau)|)\,d\tau\ll\sqrt{X}\,L\log L. \tag{62}$$

For the archimedean term, Stirling gives $|\mu(\tau)| \ll \ell$ on $|\tau| \le 3T$. Thus the part of $\mathbb{R} \setminus I$ with $|\tau| \le 3T$ is $O(\ell L \log L)$ by the same argument. If $|\tau| > 3T$, then every $\tau_k \in [T, 2T + O(h)]$ satisfies $|\tau - \tau_k| \asymp |\tau|$, and (55) gives

$$S_d(\tau) \ll d\,|\tau|^{-4} \ll TL\,|\tau|^{-4}, \qquad |\mu(\tau)| \ll \log(2+|\tau|).$$

Consequently

$$\int_{|\tau|>3T} S_d(\tau)|\mu(\tau)|\,d\tau \ll TL \int_{3T}^{\infty} \frac{\log(2+t)}{t^4}\,dt \ll \frac{L\ell}{T^2}.$$

Combining these estimates,

$$R_{\text{out}} \ll (\ell + \sqrt{X})L\log L + \frac{L\ell}{T^2} = o(L^2 N) \tag{63}$$

for $L \asymp \ell$, $X \asymp T$, since $N \asymp T\ell$.

Now the spectral representation gives the exact decomposition

$$\operatorname{tr} G = \int_I S_d(\tau)\nu_X(\tau)\,d\tau + \int_{\mathbb{R}\setminus I} S_d(\tau)\nu_X(\tau)\,d\tau.$$

Using (60), (61), and (63),

$$\operatorname{tr} G = aL^2 \int_I \nu_X(\tau)\,d\tau + o(L^2 N). \tag{64}$$

By Riemann–von Mangoldt and Stirling,

$$\int_I \mu(\tau)\,d\tau = N + O(\ell).$$

Termwise integration and (58) give

$$\left|\int_I P_X(\tau)\,d\tau\right| \ll \sum_{n\le X} \frac{\Lambda(n)}{\sqrt{n}\log n} \ll \sqrt{X} = o(N),$$

while the explicit formula for Π_X gives $\int_I \Pi_X(\tau)\,d\tau = o(N)$. Therefore

$$\operatorname{tr}\widetilde{G} = aLN(1+o(1)), \qquad \boxed{\operatorname{tr}\widehat{G} = N(1+o(1)).} \tag{65}$$

C.4 Integrated finite-grid end effects for the second moment

Define

$$K(\tau,\tau') = \sum_{0\le k<d} \widehat{\phi}(\tau-\tau_k)\widehat{\phi}(\tau'-\tau_k),$$

$$K_\infty(\tau,\tau') = \sum_{k\in\mathbb{Z}} \widehat{\phi}(\tau-\tau_k)\widehat{\phi}(\tau'-\tau_k) = L\Phi(\tau-\tau'),$$

and $K_{\text{out}} = K_\infty - K$. Since $\sum_{k\in\mathbb{Z}} |\widehat{\phi}(\tau-\tau_k)|^2 = aL^2$, Cauchy–Schwarz gives

$$|K|, |K_\infty| \le aL^2, \qquad |K_{\text{out}}(\tau,\tau')| \le \sum_{k\notin[0,d)} \psi(\tau-\tau_k)\psi(\tau'-\tau_k). \tag{66}$$

Put

$$B = C(\ell + \sqrt{X}),$$

so $|\nu_X(\tau)| \le B$ for $|\tau| \le 4T$ and $B^2 \ll \ell^2 + X$.

Lemma C.2 (Global density bound used in the tails)**.** *For* $|\tau| \le 4T$*,*

$$|\nu_X(\tau)| \ll \ell + \sqrt{X} =: B.$$

If $\tau_k \in [T, 2T]$ *and* $r = |\tau - \tau_k| > 2T$*, then*

$$|\nu_X(\tau)| \ll B + \log(r/T).$$

Proof. The prime term satisfies $|P_X| \ll \sqrt{X}$ by $\sum_{n \le X} \Lambda(n)/\sqrt{n} \ll \sqrt{X}$. The pole density satisfies $|\Pi_X(\tau)| \ll \sqrt{X}/(1+|\tau|) + 1/(1+\tau^2)$. Stirling gives $\mu(\tau) \ll 1 + \log(2+|\tau|)$. This proves the first estimate on $|\tau| \le 4T$. For $r > 2T$ and $\tau_k \in [T, 2T]$, one has $|\tau| \le r + 2T < 2r$ and hence $\log(2+|\tau|) \ll \ell + \log(r/T)$; the prime term remains bounded by $O(\sqrt{X})$ and the pole term is smaller. □

Proposition C.3 (Integrated end effects)**.** *For* $I = (T, 2T]$ *and*

$$M = \iint_{I \times I} \Phi(\tau - \tau')^2 \nu_X(\tau)\nu_X(\tau')\, d\tau d\tau',$$

one has

$$\operatorname{tr} \widetilde{G}^2 = M + O\Big(L\ell \log L\,(\ell^2 + X)\Big), \qquad \widetilde{G} = G/L. \tag{67}$$

Proof. Expanding $\operatorname{tr} G^2$ gives

$$L^2 \operatorname{tr} \widetilde{G}^2 = \iint_{\mathbb{R}^2} |K(\tau, \tau')|^2 \nu_X(\tau)\nu_X(\tau')\, d\tau d\tau'.$$

On $I \times I$, $|K_\infty|^2 = L^2\Phi^2$. Write the difference from $L^2 M$ as $E_1 + E_2$, where

$$E_1 = \iint_{I\times I} (|K|^2 - |K_\infty|^2)\nu\nu', \qquad E_2 = \iint_{(I\times I)^c} |K|^2 \nu\nu'.$$

For E_1,

$$\big||K|^2 - |K_\infty|^2\big| \le |K - K_\infty|(|K| + |K_\infty|) = |K_{\mathrm{out}}|(|K| + |K_\infty|) \ll L^2 |K_{\mathrm{out}}|.$$

Hence

$$|E_1| \ll L^2 B^2 \sum_{k \notin [0,d)} \left(\int_I \psi(\tau - \tau_k)\, d\tau\right)^2.$$

For omitted grid points at distance $\Delta > 0$ from I,

$$\int_I \psi(\tau - \tau_k) d\tau \le \int_\Delta^\infty \psi(r) dr \le \min\left\{\Psi_0, \frac{C}{\Delta}\right\}.$$

At the left edge, omitted points are $k = -j$, $j \ge 1$, and have distance jh from I. At the right edge the two exceptional omitted indices $k = d, d+1$ may lie in, or within one mesh step of, the endpoint $2T$ because $d = \lfloor T/h \rfloor$; for them we use the crude bound $2\Psi_0$. Every further right-hand omitted index has distance at least jh, $j \ge 1$. Therefore

$$\begin{aligned} S &:= \sum_{k \notin [0,d)} \left(\int_I \psi(\tau - \tau_k) d\tau\right)^2 \\ &\ll \Psi_0^2 + \sum_{j \ge 1} \min\left\{\Psi_0, \frac{C}{jh}\right\}^2. \end{aligned}$$

Let $J = \lceil C/(h\Psi_0) \rceil$. Then

$$\sum_{j \le J} \Psi_0^2 \ll \frac{\Psi_0}{h} + \Psi_0^2,$$

whereas

$$\sum_{j > J} \frac{C^2}{j^2 h^2} \ll \frac{C^2}{h^2 J} \ll \frac{\Psi_0}{h}.$$

Since $h = 2\pi/L$ and $\Psi_0 = O(\log L)$,

$$S = O(L \log L).$$

Thus

$$|E_1| \ll L^3 B^2 \log L. \tag{68}$$

For E_2, symmetry lets us place the first variable outside I and multiply by two. Since $|K|^2 \ll L^2 |K|$ and $|K| \le \sum_{0 \le k < d} \psi(\tau - \tau_k)\psi(\tau' - \tau_k)$,

$$|E_2| \ll L^2 \sum_{0 \le k < d} A_k C_k,$$

where

$$A_k = \int_{\tau \notin I} \psi(\tau - \tau_k)|\nu_X(\tau)|\, d\tau, \quad C_k = \int_{\mathbb{R}} \psi(\tau' - \tau_k)|\nu_X(\tau')|\, d\tau'.$$

We first bound C_k uniformly. On $|\tau' - \tau_k| \le 2T$ one has $|\tau'| \le 4T$, hence

$$\int_{|\tau' - \tau_k| \le 2T} \psi(\tau' - \tau_k)|\nu_X(\tau')| d\tau' \le 2B\Psi_0.$$

For $r = |\tau' - \tau_k| > 2T$, one has $|\tau'| \le 2r$ and

$$|\nu_X(\tau')| \ll B + \log(r/T), \qquad \psi(r) \ll r^{-2}.$$

Therefore the far part is

$$\ll \int_{2T}^{\infty} \frac{B + \log(r/T)}{r^2}\, dr \ll \frac{B+1}{T} \ll B,$$

and hence

$$C_k \ll B\Psi_0. \tag{69}$$

Now sum A_k before integrating. Put

$$\sigma(\tau) = \sum_{0 \le k < d} \psi(\tau - \tau_k), \qquad \Delta = \operatorname{dist}(\tau, I).$$

The distances to grid points dominate $\Delta, \Delta + h, \Delta + 2h, \ldots$, so monotonicity yields

$$\sigma(\tau) \le \psi(\Delta) + \frac{1}{h} \int_{\Delta}^{\infty} \psi(r) dr \ll \psi(\Delta) + L \min\left\{\Psi_0, \frac{C}{\Delta}\right\}. \tag{70}$$

Also, since $d \ll LT$ and $\psi(r) \ll r^{-2}$,

$$\sigma(\tau) \ll LT\, \Delta^{-2}. \tag{71}$$

For $0 < \Delta \le 2T$, $|\tau| \le 4T$, so $|\nu_X| \le B$. Integrating (70) on the two sides of I gives

$$\begin{aligned}\int_{0<\Delta\le 2T} |\nu_X(\tau)|\sigma(\tau)d\tau &\ll B\left[\Psi_0 + L\int_0^{2T} \min\left\{\Psi_0, \frac{C}{\Delta}\right\} d\Delta\right] \\ &\ll B\left[\Psi_0 + L + L\log(2T\Psi_0/C)\right] \\ &\ll BL\ell.\end{aligned}$$

For $\Delta > 2T$, use (71) and $|\nu_X(\tau)| \ll B + \log(\Delta/T)$:

$$\begin{aligned}\int_{\Delta>2T} |\nu_X(\tau)|\sigma(\tau)d\tau &\ll LT\int_{2T}^{\infty} \frac{B+\log(\Delta/T)}{\Delta^2} d\Delta \\ &\ll LB.\end{aligned}$$

Consequently

$$\sum_k A_k = \int_{\tau\notin I} |\nu_X(\tau)|\sigma(\tau)d\tau \ll BL\ell. \tag{72}$$

Combining (69) and (72),

$$|E_2| \ll L^2(B\Psi_0)(BL\ell) \ll L^3B^2\ell\log L. \tag{73}$$

Equations (68) and (73) give

$$L^2 \operatorname{tr}\widetilde{G}^2 = L^2M + O(L^3B^2\ell\log L).$$

Divide by L^2 and use $B^2 \ll \ell^2 + X$ to obtain (67). $\square$

At $L = \ell$ and $X \asymp T$, the error in (67) is $o(TL\ell^2)$.

C.5 Evaluation of the continuous second moment

For functions u_1, u_2 on I, write

$$\mathcal{M}[u_1,u_2] = \iint_{I\times I} \Phi(\tau-\tau')^2 u_1(\tau)u_2(\tau')\, d\tau\, d\tau'.$$

Then

$$M = \mathcal{M}[\mu,\mu] + \mathcal{M}[P_X,P_X] + 2\mathcal{M}[\mu,P_X] + 2\mathcal{M}[\mu,\Pi_X] + 2\mathcal{M}[P_X,\Pi_X] + \mathcal{M}[\Pi_X,\Pi_X].$$

C.5.1 Archimedean term

Using $\mu(\tau) = \mu(\tau') + O(|\tau-\tau'|/T)$, (56), and (57), one obtains

$$\mathcal{M}[\mu,\mu] = 2\pi bL\int_I \mu(\tau)^2\, d\tau + O(\ell^2\log L) = \frac{bLT\ell_1^2}{2\pi} + o(TL\ell^2). \tag{74}$$

Indeed, the local replacement error is bounded by $O(\ell\int\Phi(x)^2|x|dx)$ after integrating over I, and the truncation of the full x-integral at the two interval ends costs $O(\ell^2\int\Phi(x)^2\min(|x|,T)dx)$.

C.5.2 Prime term: diagonal and two off-diagonal families

Write

$$a_n = \frac{\Lambda(n)}{\sqrt{n}}, \qquad y_n = \log n,$$

so

$$P_X(\tau) = -\frac{1}{2\pi} \sum_{n \le X} a_n (e^{i\tau y_n} + e^{-i\tau y_n}).$$

Substitute $x = \tau - \tau'$. For $|x| < T$, the τ'-interval is $[T + x_-, 2T - x_+]$, $x_\pm = \max(\pm x, 0)$. Expanding the four exponential products splits $\mathcal{M}[P_X, P_X]$ into a quotient-frequency diagonal D, a quotient-frequency off-diagonal O_1, and a product-frequency term O_2.

For $n = m$, the inner interval has length $T - |x|$, hence Fourier inversion and (56) give

$$D = \frac{T}{\pi} \sum_{n \le X} a_n^2 g(y_n) + O(L^2 \log L). \tag{75}$$

For product frequencies,

$$\left| \int_{T+x_-}^{2T-x_+} e^{i\tau' \log(nm)}\, d\tau' \right| \le \frac{2}{\log(nm)} \le \frac{2}{\log 4}.$$

Returning to the expression before the x-integration and taking absolute values therefore gives

$$\begin{aligned} |O_2| &\le \frac{1}{2\pi^2} \sum_{n,m \le X} a_n a_m \int_{-T}^{T} \Phi(x)^2 \frac{2}{\log(nm)}\, dx \\ &\le \frac{1}{\pi^2 \log 4} \left(\sum_{n \le X} a_n \right)^2 \int_{\mathbb{R}} \Phi(x)^2\, dx. \end{aligned}$$

By the prime-power estimate

$$\sum_{n \le X} a_n = \sum_{n \le X} \frac{\Lambda(n)}{\sqrt{n}} \ll \sqrt{X}$$

and Plancherel

$$\int_{\mathbb{R}} \Phi(x)^2\, dx = 2\pi b L \ll L.$$

Consequently

$$\boxed{O_2 \ll XL.} \tag{76}$$

For $n \ne m$, put $\vartheta = y_n - y_m$ and

$$\alpha_n^+ = \int_0^T \Phi(x)^2 e^{ixy_n}\, dx, \qquad \alpha_n^- = \int_{-T}^0 \Phi(x)^2 e^{ixy_n}\, dx.$$

Then $|\alpha_n^\pm| \le \pi b L \ll L$. Direct evaluation of the inner τ'-integral gives

$$\begin{aligned} O_1 = \frac{1}{2\pi^2} \Re \sum_{n \ne m} \frac{a_n a_m}{i(y_n - y_m)} \Bigg[& \left(\frac{n}{m}\right)^{2iT} (\alpha_m^+ + \alpha_n^-) \\ & - \left(\frac{n}{m}\right)^{iT} (\alpha_n^+ + \alpha_m^-) \Bigg]. \end{aligned}$$

Thus O_1 is the sum of four Hilbert forms with coefficient pairs

$$(a_n n^{2iT}, a_m m^{-2iT}\alpha_m^+), \quad (a_n n^{2iT}\alpha_n^-, a_m m^{-2iT}),$$
$$(a_n n^{iT}\alpha_n^+, a_m m^{-iT}), \quad (a_n n^{iT}, a_m m^{-iT}\alpha_m^-).$$

We use the weighted Montgomery–Vaughan Hilbert inequality

$$\left|\sum_{r\neq s}\frac{x_r z_s}{\lambda_r-\lambda_s}\right| \ll \left(\sum_r \frac{|x_r|^2}{\delta_r}\right)^{1/2}\left(\sum_r \frac{|z_r|^2}{\delta_r}\right)^{1/2}, \quad \delta_r = \min_{s\neq r}|\lambda_r-\lambda_s|,$$

with an absolute implied constant; no sharp constant is needed. For $\lambda_n = \log n$, $\delta_n^{-1} \ll n$, since adjacent integers already have logarithmic spacing $\gg 1/n$. Hence

$$\sum_{n\le X}\frac{a_n^2}{\delta_n} \ll \sum_{n\le X}\Lambda(n)^2 \ll XL,$$

and a sequence carrying $\alpha_n^\pm$ has the corresponding weighted square sum $\ll L^3 X$. Each of the four Hilbert forms is therefore $O(L^2X)$, so

$$O_1 \ll L^2 X. \tag{77}$$

Combining (75)–(77),

$$\mathcal{M}[P_X, P_X] = \frac{T}{\pi}\sum_{n\le X}\frac{\Lambda(n)^2}{n}g(\log n) + O(L^2X). \tag{78}$$

By Stieltjes partial summation from (59), and using $\|g'\|_\infty \le \|(\phi^2)'\|_1 = O(1)$,

$$\sum_{n\le X}\frac{\Lambda(n)^2}{n}g(\log n) = \int_0^L y g(y)\,dy + O(L^2). \tag{79}$$

C.5.3 Mixed and pole terms

Proposition C.4 (Secondary terms)**.** *At $L=\ell$ and $X \asymp T$,*

$$\mathcal{M}[\mu, P_X] \ll \ell\sqrt{X}, \qquad \mathcal{M}[\mu, \Pi_X] \ll \ell L\sqrt{X},$$
$$\mathcal{M}[P_X, \Pi_X] \ll LX, \qquad \mathcal{M}[\Pi_X, \Pi_X] \ll \frac{LX}{T}.$$

All four are $o(TL\ell^2)$.

Proof. Define

$$m(\tau') = \int_I \Phi(\tau-\tau')^2\mu(\tau)\,d\tau.$$

Then $\|m\|_\infty \ll \ell L$. Differentiating under the integral and integrating by parts in τ gives

$$\int_I |m'(\tau')|\,d\tau' \ll \ell L,$$

using $\mu' = O(T^{-1})$ and (57). Therefore, for $y \ge \log 2$,

$$\left|\int_I m(\tau')e^{iy\tau'}d\tau'\right| \ll \frac{\ell L}{y}.$$

Substitution of the defining formula for P_X and $\sum_{n\le X}\Lambda(n)/(\sqrt{n}\log n) \ll \sqrt{X}/L$ yields $\mathcal{M}[\mu, P_X] \ll \ell\sqrt{X}$.

For the remaining terms use on I

$$|\Pi_X| \ll \frac{\sqrt{X}}{T}, \qquad 0<\mu\ll\ell, \qquad |P_X|\ll\sqrt{X},$$

and

$$\sup_\tau \int_I \Phi(\tau-\tau')^2 d\tau' \le 2\pi bL \ll L.$$

Thus

$$|\mathcal{M}[\mu,\Pi_X]| \ll T\cdot\ell\cdot\frac{\sqrt{X}}{T}\cdot L = \ell L\sqrt{X},$$
$$|\mathcal{M}[P_X,\Pi_X]| \ll T\cdot\sqrt{X}\cdot\frac{\sqrt{X}}{T}\cdot L = LX,$$

and

$$|\mathcal{M}[\Pi_X,\Pi_X]| \ll T\cdot\frac{X}{T^2}\cdot L = \frac{LX}{T}.$$

□

C.6 Scale-free limit

Assume the fixed-width tapered family satisfies

$$\phi_L(Ls)^2 = v(s) + o(1)$$

in $L^1([-1/2,1/2])$, with v bounded and supported on $[-1/2,1/2]$. Then

$$a \to A := \int_{-1/2}^{1/2} v(s)\,ds, \qquad b \to B_v := \int_{-1/2}^{1/2} v(s)^2\,ds.$$

Writing

$$G_v(r) = \int v(s)v(s-r)\,ds,$$

one has $g(Lr) = LG_v(r) + o(L)$ in L^1, and hence

$$\frac{2}{L^3}\int_0^L y g(y)\,dy \to J_v := \iint_{[-1/2,1/2]^2} |s-t| v(s)v(t)\,ds\,dt. \tag{80}$$

Combining Proposition C.3, (74), (78), (79), Proposition C.4, and (80),

$$\operatorname{tr}\widetilde{G}^2 = \frac{TL}{2\pi}\left(B_v\ell_1^2 + L^2 J_v\right) + o(TL\ell^2).$$

Since, by (50),

$$N(T,2T) = \frac{T\ell_1}{2\pi} + O(\ell), \qquad \frac{L}{\ell_1}\to 1,$$

and $\widehat{G} = \widetilde{G}/(aL)$,

$$\boxed{\|\widehat{G}\|_{\mathrm{F}}^2 = \left(\frac{B_v+J_v}{A^2} + o(1)\right) N(T,2T).} \tag{81}$$

Thus

$$c_1(v) = \frac{A^2}{B_v+J_v}.$$

Together with (65), this proves the two required prime-side moments for every admissible profile.

C.7 Closing the Montgomery–Taylor optimization

On $H = L^2([-1/2, 1/2])$ define the self-adjoint integral operator

$$(\mathcal{T}v)(s) = \int_{-1/2}^{1/2} |s-t| v(t)\, dt.$$

Then

$$c_1(v) = \frac{\langle 1, v\rangle^2}{\langle v, (I+\mathcal{T})v\rangle}.$$

Lemma C.5 (Positivity and invertibility)**.** *$I + \mathcal{T}$ is strictly positive and invertible on H.*

Proof. Schur's test gives

$$\|\mathcal{T}\|_{2\to 2} \le \sup_{|s|\le 1/2} \int_{-1/2}^{1/2} |s-t|\, dt = \sup_{|s|\le 1/2} \left(s^2 + \frac{1}{4}\right) = \frac{1}{2}.$$

Hence

$$\langle v, (I+\mathcal{T})v\rangle \ge (1 - \|\mathcal{T}\|)\|v\|_2^2 \ge \frac{1}{2}\|v\|_2^2.$$

$\square$

Cauchy–Schwarz in the positive inner product $\langle f, g\rangle_A = \langle f, (I+\mathcal{T})g\rangle$ now gives

$$c_1(v) \le \langle 1, (I+\mathcal{T})^{-1} 1\rangle,$$

with equality precisely when

$$v \propto (I+\mathcal{T})^{-1} 1.$$

Thus a maximizer satisfies

$$(I+\mathcal{T})v = C \tag{82}$$

for some constant C. In the interior,

$$(\mathcal{T}v)'' = 2v,$$

so (82) implies

$$v'' + 2v = 0.$$

By symmetry the maximizing solution is even, hence

$$v(s) = A_0 \cos(\sqrt{2}s).$$

Conversely, for $v_*(s) = \cos(\sqrt{2}s)$ the function $F = v_* + \mathcal{T}v_*$ satisfies $F'' = 0$ and is even, hence is constant; therefore v_* really satisfies (82), not merely its differentiated equation. Since $1/\sqrt{2} < \pi/2$, $v_* > 0$ on the whole interval, so the constraint $v \ge 0$ is inactive.

Let $\theta = 1/\sqrt{2}$. Then

$$A = \int_{-1/2}^{1/2} v_*(s)\, ds = \sqrt{2}\sin\theta.$$

Evaluating $C = (I+\mathcal{T})v_*$ at $s = 0$ gives

$$C = 1 + 2\int_0^{1/2} t\cos(\sqrt{2}t)\, dt = \cos\theta + \frac{\sin\theta}{\sqrt{2}}.$$

Since $\langle v_*, (I+\mathcal{T})v_*\rangle = CA$,
$$c_1^* = \frac{A}{C} = \frac{2\tan(1/\sqrt{2})}{\sqrt{2}+\tan(1/\sqrt{2})},$$
and therefore
$$\boxed{\frac{1}{c_1^*} = \frac{1}{2} + \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}}.} \tag{83}$$

D Independent proof of Input IV: Poisson–Gabor sampling

This appendix proves the critical-lattice Poisson identity, the tapered Fourier decay, finite-grid truncation, and the Montgomery–Taylor overlap limit. The window is real and even, so the real-axis transforms used below are real and even; modulus squares may be retained throughout.

D.1 Statement to be verified

Let $L \to \infty$, put
$$h = \frac{2\pi}{L}, \qquad \tau_k = kh,$$
and let $\phi_L \in C_c^2(\mathbb{R})$ be a real tapered window with exact support $[-L/2, L/2]$. Write
$$\widehat{\phi}_L(\xi) = \int_{\mathbb{R}} \phi_L(u) e^{-i\xi u}\, du$$
and
$$a_L = \frac{1}{L}\int_{\mathbb{R}} \phi_L(u)^2\, du.$$
For the Montgomery–Taylor choice, away from an $O(1)$ physical taper layer,
$$\phi_L(u)^2 = \cos\left(\frac{\sqrt{2}\,u}{L}\right), \qquad |u| \le \frac{L}{2} - O(1).$$
Define
$$K(x) = \int_{-1/2}^{1/2} \cos(\sqrt{2}\,t)\cos(2\pi x t)\, dt, \qquad k(x) = \frac{K(x)}{K(0)}.$$

The required form of Input IV is the following.

Theorem D.1 (Independent Input IV)**.** *Fix $R > 0$. Let γ, γ' be points whose normalized coordinates $x = L\gamma/(2\pi)$ and $x' = L\gamma'/(2\pi)$ are at least L^2 grid cells from both ends of the finite sampling interval, and suppose $|x - x'| \le R$. Then the normalized finite-grid Gram entry satisfies*
$$\frac{1}{a_L L^2} \sum_{\tau_j \text{ in the finite grid}} \widehat{\phi}_L(\gamma - \tau_j)\overline{\widehat{\phi}_L(\gamma' - \tau_j)} = k(x - x') + o(1),$$
uniformly for $|x - x'| \le R$.

D.2 The infinite-grid identity

Lemma D.2 (Critical-grid Poisson identity)**.** *Assume $\phi \in L^2(\mathbb{R})$ is compactly supported in an interval of length at most L and is sufficiently regular for Poisson summation. Then*

$$\sum_{k\in\mathbb{Z}} \widehat{\phi}(\tau-\tau_k)\overline{\widehat{\phi}(\tau'-\tau_k)} = L\int_{\mathbb{R}} |\phi(u)|^2 e^{-i(\tau-\tau')u}\,du.$$

Proof. Set

$$F_{\tau,\tau'}(\xi) = \widehat{\phi}(\tau-\xi)\,\overline{\widehat{\phi}(\tau'-\xi)}.$$

With the Fourier convention above, a direct Fourier inversion calculation gives

$$\widehat{F}_{\tau,\tau'}(s) = 2\pi e^{-i\tau' s}\int_{\mathbb{R}} \phi(u)\overline{\phi(u-s)}e^{-i(\tau-\tau')u}\,du.$$

Poisson summation on the lattice $h\mathbb{Z}$, $h = 2\pi/L$, gives

$$\sum_{k\in\mathbb{Z}} F_{\tau,\tau'}(kh) = \frac{1}{h}\sum_{m\in\mathbb{Z}} \widehat{F}_{\tau,\tau'}(mL).$$

Because ϕ is supported in an interval of length at most L, the translated supports of $\phi(u)$ and $\phi(u-mL)$ are disjoint for $m \neq 0$, apart from endpoints of measure zero. Thus only $m = 0$ survives. Since $1/h = L/(2\pi)$,

$$\sum_{k\in\mathbb{Z}} F_{\tau,\tau'}(kh) = L\int_{\mathbb{R}} |\phi(u)|^2 e^{-i(\tau-\tau')u}\,du.$$

□

Remark D.3. *The exact support condition* $\operatorname{supp}\phi_L = [-L/2, L/2]$ *is the one used by the optimized fixed-width tapered family. Hence the critical-density Poisson identity has no aliasing term: shifts by mL, $m \neq 0$, overlap only on sets of measure zero.*

D.3 Fourier decay

Lemma D.4 (Uniform decay)**.** *Suppose $\phi_L \in C_c^2([-L/2, L/2])$ is uniformly bounded and its taper has fixed physical width with uniformly bounded first and second derivatives. Then*

$$|\widehat{\phi}_L(r)| \ll \min\left\{L, \frac{1}{|r|}, \frac{1}{r^2}\right\},$$

with an implied constant independent of L.

Proof. The L^1 estimate gives $|\widehat{\phi}_L(r)| \ll L$. One integration by parts, using compact support and the bounded total variation of ϕ_L, gives $|\widehat{\phi}_L(r)| \ll |r|^{-1}$. A second integration by parts on each smooth piece gives $|\widehat{\phi}_L(r)| \ll |r|^{-2}$: the bulk derivatives of the Montgomery–Taylor profile contribute $O(L^{-1})$ and $O(L^{-2})$ per unit length, while the fixed-width taper contributes $O(1)$ total variation at each derivative level. All boundary terms vanish because the taper is compactly supported and C^2. □

D.4 Finite-grid truncation

Lemma D.5 (Pointwise tail)**.** *Fix $R > 0$. Suppose γ, γ' are at least L^2 grid cells from a finite-grid endpoint and $|L(\gamma - \gamma')/(2\pi)| \le R$. Then the contribution of the omitted half-grid beyond that endpoint is $O(L^{-2})$ before Gram normalization.*

Proof. Let the omitted index be $n \ge L^2$ cells beyond the nearer endpoint. Since $h = 2\pi/L$ and the normalized separation of γ, γ' is bounded,

$$|\gamma - \tau_j| \asymp \frac{n}{L}, \qquad |\gamma' - \tau_j| \asymp \frac{n}{L}$$

uniformly in the stated region. Lemma D.4 therefore gives

$$|\widehat{\phi}_L(\gamma - \tau_j)\widehat{\phi}_L(\gamma' - \tau_j)| \ll \frac{L^4}{n^4}.$$

Consequently

$$\sum_{n \ge L^2} \frac{L^4}{n^4} \ll L^4 \int_{L^2-1}^{\infty} t^{-4}\, dt = O(L^{-2}).$$

The same estimate applies at the other endpoint. □

D.5 Montgomery–Taylor overlap limit

Lemma D.6 (Normalization and overlap)**.** *For the fixed-width tapered Montgomery–Taylor family,*

$$a_L = K(0) + o(1), \qquad K(0) = \int_{-1/2}^{1/2} \cos(\sqrt{2}t)\, dt = \sqrt{2}\sin(1/\sqrt{2}) > 0.$$

Moreover, uniformly for $|x| \le R$,

$$\frac{1}{a_L L} \int_{\mathbb{R}} \phi_L(u)^2 e^{-i(2\pi x/L)u}\, du = k(x) + o(1).$$

Proof. Put $u = Lt$. Outside a set of t-measure $O(L^{-1})$ arising from the fixed-width taper,

$$\phi_L(Lt)^2 = \cos(\sqrt{2}t)$$

on $[-1/2, 1/2]$. Uniform boundedness of the window makes the taper contribution $O(L^{-1})$ after normalization. Dominated convergence, uniformly for $|x| \le R$, gives

$$a_L = \int_{-1/2}^{1/2} \cos(\sqrt{2}t)\, dt + o(1) = K(0) + o(1)$$

and

$$\frac{1}{L} \int \phi_L(u)^2 e^{-i(2\pi x/L)u}\, du = \int_{-1/2}^{1/2} \cos(\sqrt{2}t) e^{-i2\pi xt}\, dt + o(1).$$

The odd imaginary part vanishes. Dividing by $a_L = K(0) + o(1)$ proves the claim. □

The overlap has the explicit form

$$K(x) = \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}},$$

with removable singularities interpreted by continuity.

Proof of Theorem D.1. Apply Lemma D.2 to the infinite grid. By Lemma D.5, replacing the infinite grid by the finite central grid changes the unnormalized sum by $O(L^{-2})$. Since $a_L \asymp 1$, division by $a_L L^2$ makes this error $O(L^{-4})$. The remaining infinite-grid expression, after normalization, is the overlap in Lemma D.6. This gives $k(x - x') + o(1)$ uniformly on every fixed compact set of normalized separations. □

D.6 Audit notes and exact scope

The proof above is independent of the zeta explicit formula, zero density, and prime-side pair correlation. It uses only:

1. Poisson summation on the critical lattice $2\pi\mathbb{Z}/L$;
2. compact support and regularity of the chosen window;
3. the fixed-width taper;
4. elementary integration by parts.

Thus Input IV is proved independently from Poisson summation and elementary Fourier estimates for the exact support/taper convention used in the main manuscript.

E Certificate metadata

The frozen archival verifier is

`certify_nonuniform_3900_v1_1.py`.

Its SHA-256 is
`8eb7b4a17da8e114881264d3c2fc8daf262a0558d4248692b386c0fca2cc4301`.
The 256-bit log SHA-256 is
`4b0e62dcb5c4014504981f16e431144e3a01184b2aa2aa6b0bf650705d59db83`.

Field	Archived 256-bit run
verified	`true`
target	$F_{6,\text{nonuniform}} \geq 39/10000$
grid	4000
precision	256 bits
nodes	1,119,372
pruned	560,334
splits	559,038
maximum depth	39
initial boxes	1296
interval prunes	364,112
tangent prunes	190,953
pressure prunes	5,269
elapsed time	201.487482 s

The pressure vector recorded by the verifier is

$$(2714, 3733, 3553, 3553, 3733, 2714)/10^7.$$

The kernel-table SHA-256 is
`f75a3968f6daf7ea74867e430dc69fdca4276c4bb09f8cecc1b9846e8942c3f5.`

The second-derivative-table SHA-256 is
`f878d03d2eeca7fd34e3064a8dacb70d054fe3b3e9e3890c79a1f8297633cfb3.`

The archived environment reports CPython 3.14.7 and `python-flint==0.9.0`.

F Exact arithmetic

From $m = 262$ and $A_0 = 624/625$,

$$\frac{A_0}{m} = \frac{312}{81875}, \qquad \frac{m-1}{500m} = \frac{261}{131000}, \qquad 1 - \frac{312}{81875} = \frac{81563}{81875}.$$

Thus

$$\frac{H_{\mathrm{MT}} - 261/131000}{81563/81875} = \frac{655000H_{\mathrm{MT}} - 1305}{652504}.$$

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